

Generated Gates Inherit Their Generator’s Blind Spots: A Diversity Floor for Self-Generated Verification

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Abstract

A growing body of empirical work reports that when language models generate the artifacts that check other language models—unit tests, verifiers, evaluators, monitors—the checkers inherit the generator’s failure modes: machine-generated test suites mirror the generating model’s error patterns and miss the errors a human would make, model evaluators recognize and favor their own outputs, and the self-preference extends to architecturally related model *families*. We give a representation-theoretic account of this phenomenon and prove two limits. Modeling a gate by the indistinguishability relation of its underlying model family F , with a *stealth set* Σ_F of faults the family cannot separate from acceptable artifacts, we show **(GG1, inheritance)** that any gate constructed within F has a blind set containing Σ_F —it cannot do better than the family it is built from—and **(GG2, the self-generation floor)** that a closed system that generates its own gates from a single founding family has an escape floor exactly equal to its stealth mass $\lambda_F = \mu(\Sigma_F)$ (its blind set stabilizing at Σ_F), a fixed point invariant to generation depth, recursion, and prompt diversity. The floor is strictly lowerable only by an *exogenous* distinction source: a genuinely different family, whose independent stealth set multiplies the floor down, or a reality-grounded gate that consults ground truth rather than any family’s representation and is the unique non-representational escape. The results generalize the classical theory of coincident failures in N -version programming [1, 2]—which assumed independently developed, *exogenous* versions—to the regime where the versions and their checkers are produced from within one representational family, a regime that theory did not address; the new content is that diversity cannot be bootstrapped from inside a family. We situate the floor as the origin of the stealth mass that bounds verification in long-horizon agentic work [17], illustrate all claims in simulation, and state what must be measured to make the floor quantitative for a given system.

1 Introduction

The dominant strategy for making unreliable generative systems usable is to check their outputs: generate a candidate, then accept it only if it passes one or more gates—tests, static checks, a second model asked to find fault, a panel of model judges. When the gates are good and disagree in their mistakes, this works, and works quantitatively: a stack of imperfect but diverse gates can drive the probability that a faulty artifact is accepted to an arbitrarily small value, and can extend the reliable horizon of an unreliable executor without bound at overhead logarithmic in the horizon [17]. The premise that does the work is *diversity*: the gates must not share the faults they miss.

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It has become common to generate the gates the same way the candidates are generated—to write the unit tests with the same model that wrote the code, to ask a model to critique its own output, to build an evaluator panel out of one model family. The empirical record on this is now consistent and unflattering. Machine-generated test suites exhibit tightly clustered error patterns and a “homogenization trap,” detecting the failures a model is prone to while missing the diverse mistakes a human would make [10]; verifiers built from such suites carry the resulting blind spots and impose a “verification ceiling” on what they can certify [11]; model evaluators recognize their own generations and score them higher than human judges do, an effect causally tied to self-recognition [8] and traced to a familiarity preference for low-perplexity, self-like text, and the bias extends from the individual model to its architectural *family* [9]; and even heterogeneous multi-agent teams have been observed to fail to match their single best member, with homogeneous copies making matters worse [12]; and same-family model ensembles collapse to a handful of effective voters, architectural kin sharing the very errors a calibrated vote then amplifies [13]. Each of these is an observation. What has been missing is an account of *why* it is a hard limit rather than a tendency to be engineered away, and of exactly what *can* escape it.

This paper supplies that account. We adopt the indistinguishability model that underlies the diversity premise: a gate built from a model family can only act on distinctions that family can represent, and is therefore blind to the faults that family confuses with acceptable artifacts. We call that confusable-fault set the family’s *stealth set* Σ_F and prove two facts.

- **Inheritance (Theorem 1).** Any gate constructed within family F has a blind set that contains Σ_F . A generated gate cannot resolve a fault its generator cannot resolve; it inherits the generator’s blind spots.
- **The self-generation floor (Theorem 2).** A closed system that generates its own gates from a single founding family F —to any depth, with gates checking gates, under any amount of prompt variation—has an escape floor exactly $\lambda_F = \mu(\Sigma_F)$. The floor is a fixed point of the self-generation operator: it cannot be lowered from inside the family.

The floor is escapable, but only from outside. A gate from a genuinely different family N has its own stealth set $\Sigma_N \neq \Sigma_F$; running it alongside the F -stack leaves only the faults blind to both, of mass $\approx \lambda_F \lambda_N$ when the families are independent (Proposition 1). A *reality-grounded* gate—one that decides by consulting ground truth, e.g. by executing the candidate against a real oracle, rather than by any family’s representation—can reject faults in Σ_F on the scope its grounding covers, and is the unique escape that is not itself a family (Proposition 2). Neither route is available to a closed single-family system, which is precisely the content of the floor.

Relation to prior theory. The inevitability of correlated failure among independently produced versions of a program is a classical result. Eckhardt and Lee [1] showed that independent *development* does not yield independent *failure*: because all versions are derived from one specification, there is a nonzero set of inputs misinterpreted in a common way, and a “difficulty function” over the input space induces positive failure correlation. Littlewood and Miller [2] extended this to *forced* (methodological) diversity, proving that diverse development methods can yield better-than-independent behavior. Knight and Leveson [3, 4] demonstrated the correlation empirically, as did the PODS experiment [5]; Hatton [6] argued a single good version often outperforms N diverse ones, and most recently N -version units built from coding agents were found to fail coincidentally far above the independence prediction [7]. This theory was built for, and assumes, versions developed by independent *exogenous* agents—separate teams, separate methodologies. Our setting is the one that theory did not consider: the versions, and the gates that check them, are produced from

within a single representational family, and the question is whether such a system can supply itself the diversity Littlewood and Miller require. Theorem 2 answers no. We recover the classical results as the exogenous limit—a different family is methodological diversity realized in representation space, and its effect is exactly the better-than-independent multiplication of Proposition 1—and add the impossibility of obtaining that diversity endogenously. A companion analysis [17] shows that a shared stealth mass of size λ_{st} caps the reliable horizon of gated long-horizon work at $\approx 1/\lambda_{\text{st}}$ regardless of gate count; the present paper identifies where that stealth mass comes from, why generated gates carry it, and what reduces it (Remark 4).

Two recent results bound self-correction by other routes. A diagonalization argument shows that every enumerable model fails on some input, so a model correcting its own family inherits that existence limit [15]; an information-theoretic bound constrains safety verification of self-improving systems from finite observation [16]. Our floor is complementary to and orthogonal to both: a *quantified* bound at the founding family’s stealth *mass*, located in a family’s representation rather than in computability or sample complexity, and—unlike an existence or capacity limit—explicitly lowerable, by exogenous family diversity or reality-grounding.

2 Model

Fix a task with a ground-truth correctness predicate φ on a space of candidate artifacts $\mathcal{C}$. Write $\mathcal{C}_1 = \{c : \varphi(c) = 1\}$ for the acceptable candidates and $\mathcal{C}_0 = \{c : \varphi(c) = 0\}$ for the faulty ones. A faulty candidate that is accepted and promoted is said to *escape*. We measure faults with a fixed distribution μ on $\mathcal{C}_0$ (the operative fault-generating distribution); all masses below are μ -masses, and “=” between fault sets may be read up to μ -null sets.

Definition 1 (Gate and blind set). A *gate* g is a decision procedure assigning each candidate a verdict in $\{\text{accept}, \text{reject}\}$. Its *blind set* is the faulty mass it accepts, $\mathcal{B}(g) = \{c \in \mathcal{C}_0 : g(c) = \text{accept}\}$. A stack of gates $G = \{g_1, \dots, g_k\}$ accepts a candidate iff every gate accepts it, so a fault escapes the stack iff it lies in $\bigcap_i \mathcal{B}(g_i)$; the stack’s *escape mass* is $\mathbf{E}(G) = \mu(\bigcap_i \mathcal{B}(g_i))$.

We require gates to be usable, i.e. to admit acceptable work: a gate has *false-reject mass* $\rho(g) = \nu\{c \in \mathcal{C}_1 : g(c) = \text{reject}\}$ under a distribution ν on $\mathcal{C}_1$, and we restrict attention to gates with $\rho(g)$ below a fixed tolerance ρ_0 . (A gate that rejects everything has empty blind set but is useless; the tolerance excludes it.)

Definition 2 (Family and stealth set). A model *family* F has a fixed, finite parameter set; we characterize it by a *representation map* r_F on $\mathcal{C}$ recording everything the family computes about a candidate *across all of its admissible prompt and scaffold configurations*, at F ’s resolution. It induces the relation $c \sim_F c' \iff r_F(c) = r_F(c')$: c and c' are indistinguishable to F under *every* prompt. As the kernel of a map, $\sim_F$ is an equivalence relation (in particular transitive), and “ F cannot distinguish c from c' ” means exactly $c \sim_F c'$. A gate is *built within* F , or an *F -gate*, if its verdict is a function of the $\sim_F$ -class of its input: it can only condition on what F can represent. The *stealth set* of F is

$$\Sigma_F = \{c \in \mathcal{C}_0 : \exists c' \in \mathcal{C}_1 \text{ with } c \sim_F c'\},$$

the faulty candidates that F cannot separate from some acceptable candidate, with stealth mass $\lambda_F = \mu(\Sigma_F)$.

The relation $\sim_F$ is the formal carrier of “what the family can tell apart.” Because r_F ranges over all admissible prompt and scaffold configurations of a *fixed* parameter set, Definition 2 makes a gate’s

discriminating power a property of its substrate, not of the prompt wrapped around it: in-context learning and few-shot scaffolds navigate this fixed capacity—reweighting what the parameters already compute—but cannot add a representational dimension the parameters and training do not support, and Σ_F is exactly the intersection of the per-prompt blind spots over all such configurations. This is the model assumption that does the work, and it is the formal content of the empirical observation that self-preference is a family-level, representation-level effect [9, 8] rather than a surface artifact.

Remark 1 (Deterministic reduction of stochastic gates). A stochastic gate—an LLM judge at temperature $T > 0$ —need not return a single verdict per candidate. We take its verdict to be the deterministic reduction (greedy $T \rightarrow 0$ decoding, or the thresholded expectation $\mathbf{1}[\Pr(\text{accept} \mid r_F(c)) \geq \frac{1}{2}]$), which is a function of $r_F(c)$ and to which GG1–GG2 apply. Sampling variation at $T > 0$ is then an additional false-accept/false-reject noise source, folded into the tolerances and orthogonal to the representational blind set: re-rolling a verdict cannot manufacture a distinction r_F lacks, so it cannot reject a fault in Σ_F .

3 Inheritance

Theorem 1 (GG1: generated gates inherit the generator’s stealth). *Let g be a sound F -gate: $\rho(g) = 0$ under a distribution ν that assigns positive mass to every $\sim_F$ -class meeting $\mathcal{C}_1$. Then $\mathcal{B}(g) \supseteq \Sigma_F$, and consequently $\mathbf{E}(\{g\}) \geq \lambda_F$. More generally, suppose a uniform likelihood-ratio bound holds: pushing μ and ν forward to the space of $\sim_F$ -classes (the faulty and good mass each class carries) as $\bar{\mu}, \bar{\nu}$, one has $d\bar{\mu}/d\bar{\nu} \leq \kappa$ on $\text{supp } \bar{\nu}$ for a finite global κ . Then for $\rho(g) \leq \rho_0$, $\mathcal{B}(g) \supseteq \Sigma_F$ except on a stealth set of mass at most $\kappa\rho_0$, which vanishes as $\rho_0 \rightarrow 0$; absent such a finite κ , only the sound case is claimed.*

Proof. Let $c \in \Sigma_F$. By definition there is $c' \in \mathcal{C}_1$ with $c \sim_F c'$, so $g(c) = g(c')$ because g ’s verdict is a function of the $\sim_F$ -class. If $g(c') = \text{accept}$ then $g(c) = \text{accept}$ and $c \in \mathcal{B}(g)$; the only way $c \notin \mathcal{B}(g)$ is $g(c') = \text{reject}$, i.e. g rejects the good part of c ’s class. For a sound gate ($\rho = 0$) no good class of positive ν -mass is rejected, so no such c exists and $\Sigma_F \subseteq \mathcal{B}(g)$. For $\rho(g) \leq \rho_0$, the rejected good classes carry total $\bar{\nu}$ -mass at most ρ_0 ; by $d\bar{\mu}/d\bar{\nu} \leq \kappa$ their paired stealth parts carry $\bar{\mu}$ -mass at most $\kappa\rho_0$, so $\Sigma_F \subseteq \mathcal{B}(g)$ off a set of mass $\leq \kappa\rho_0$. The escape bound follows since $\mathcal{B}(g) \supseteq \Sigma_F$ gives $\mu(\mathcal{B}(g)) \geq \lambda_F$. $\square$

Remark 2 (Generator versus substrate). Theorem 1 concerns the family on whose representation the gate *decides*. A model of family F may *author* a gate whose decisions are made by a different mechanism—a second model, or an external tool—in which case the operative family is the deciding mechanism’s, not the author’s. Two cases matter and are treated below: the deciding mechanism is another model family (Proposition 1) or it is a ground-truth oracle (Proposition 2). A gate both authored and decided within F —a checklist F writes and F applies, a model grading its own family’s work—is the case Theorem 1 governs, and is the case that arises by default when a system generates its own checkers.

4 The Self-Generation Floor

We now consider a system that produces its own gates. A *closed self-generating verification system rooted in F* is any collection of gates obtained as follows: an initial set of F -gates, together with any further gates produced by generators whose own decisions factor through $\sim_F$, closed under

such generation to arbitrary depth. The defining restriction is that no exogenous representation enters: every gate in the closure, and every generator that produced it, is an F -gate in the sense of Definition 2. Let $\mathcal{G}_F$ denote the (possibly infinite) set of gates reachable in such a closure, and let its escape floor be $E_F^* = \inf\{E(G) : G \subseteq \mathcal{G}_F\}$, the least faulty mass any sub-stack drawn from the closure can leave accepted.

Theorem 2 (GG2: the self-generation floor is a fixed point at λ_F). *For any closed self-generating verification system rooted in F ,*

$$E_F^* = \lambda_F.$$

The floor is invariant to generation depth, to gates checking gates, and to prompt diversity: enlarging the closure by any amount of within- F generation does not lower it. Equivalently, the self-generation operator that maps a stack to any stack augmented with further F -gates fixes the blind set at Σ_F , hence the floor at λ_F : it can neither go below it nor, with sufficient within- F coverage, fail to reach it.

Proof. Lower bound $E_F^* \geq \lambda_F$. A stack rejects a fault iff at least one of its gates rejects it. By Theorem 1, every gate $g \in \mathcal{G}_F$ has $\mathcal{B}(g) \supseteq \Sigma_F$ (for sound gates; off a set of mass $\leq \kappa\rho_0$ otherwise), so no gate in the closure rejects any fault in Σ_F . Hence $\Sigma_F \subseteq \bigcap_{g \in G} \mathcal{B}(g)$ for every $G \subseteq \mathcal{G}_F$, giving $E(G) \geq \lambda_F$ and therefore $E_F^* \geq \lambda_F$.

Upper bound (achievability) $E_F^* \leq \lambda_F$. Outside Σ_F , every fault $c \in \mathcal{C}_0 \setminus \Sigma_F$ is, by Definition 2, $\sim_F$ -separated from all of $\mathcal{C}_1$: its $\sim_F$ -class contains no acceptable candidate. An F -gate can therefore reject c without any false reject, and some F -gate does (a gate that rejects exactly the all-faulty $\sim_F$ -classes is an F -gate with $\rho = 0$). Prompt and instance variation within F ranges over such admissible gates; a closure that includes, for each all-faulty class, a gate rejecting it has $\bigcap_g \mathcal{B}(g) = \Sigma_F$ and hence $E(G) = \lambda_F$. Thus the floor is attained.

Fixed point. The two bounds give $E_F^* = \lambda_F$. Augmenting any $G \subseteq \mathcal{G}_F$ with further F -gates cannot remove Σ_F from the intersection (lower bound) and cannot reduce the floor below the already-attained λ_F (it is the infimum); the floor is therefore stationary under within- F generation. Depth, gates-checking-gates, and prompt diversity are all instances of within- F generation, since a generator whose decisions factor through $\sim_F$ produces only $\sim_F$ -measurable gates regardless of how it is prompted or composed. $\square$

Remark 3 (Why “trying harder” does not help). Theorem 2 is the formal statement that one cannot escape a family’s blind spot by using more of the same family. Generating a thousand tests with a model, having a model critique its own critique, building a deeper panel of judges of one lineage, or searching over prompts—each is within- F generation and each leaves Σ_F exactly where it was. This is the limit behind the empirically reported degeneration of single-model self-review into repetition of the original error [10, 12], now as a consequence rather than an observation.

5 Escaping the Floor

The floor is a statement about closed single-family systems. Two distinct moves leave the closure and lower it.

Proposition 1 (Exogenous family: independent stealth multiplies the floor). *Let N be a family with stealth set Σ_N and let a stack consist of an F -component that rejects all faults outside Σ_F together with one N -gate that rejects all faults outside Σ_N . The stack’s escape mass is $\mu(\Sigma_F \cap \Sigma_N)$. If the families’ blind spots are independent under μ , $\mu(\Sigma_F \cap \Sigma_N) = \lambda_F \lambda_N$; in general $\mu(\Sigma_F \cap \Sigma_N) \leq$*

$\min(\lambda_F, \lambda_N)$, with strict reduction below λ_F whenever Σ_N does not contain Σ_F . With k families whose stealth sets are mutually independent, the floor is $\prod_{i=1}^k \lambda_i$.

Proof. A fault escapes the stack iff every gate accepts it; the F -component accepts exactly Σ_F and the N -gate accepts exactly Σ_N , so the escaping set is $\Sigma_F \cap \Sigma_N$ with mass $\mu(\Sigma_F \cap \Sigma_N)$. Independence gives the product; the bound $\mu(\Sigma_F \cap \Sigma_N) \leq \min(\lambda_F, \lambda_N)$ is monotonicity of measure; and if $\Sigma_F \not\subseteq \Sigma_N$ then $\mu(\Sigma_F \cap \Sigma_N) < \lambda_F$. The k -family statement is the same argument on $\bigcap_i \Sigma_i$ under mutual independence. $\square$

A different family is methodological diversity [2] realized as representational diversity, and Proposition 1 is its better-than-independent payoff. That payoff is the *independent* case $\mu(\Sigma_F \cap \Sigma_N) = \lambda_F \lambda_N$; independence of blind spots across families is itself a strong, empirically checkable hypothesis—precisely the one Eckhardt and Lee [1] showed independent *development* does not by itself deliver—so it is the joint mass $\mu(\Sigma_F \cap \Sigma_N)$, not the product, that a deployment must measure (§7). The essential point for the floor is the hypothesis: N must be *exogenous*, $\Sigma_N \neq \Sigma_F$. By Theorem 2 a closed F -system cannot manufacture such an N ; every gate it generates has stealth $\supseteq \Sigma_F$ and contributes nothing to $\Sigma_F \cap \Sigma_N$. Diversity that lowers the floor must be imported, not bred.

Proposition 2 (Reality-grounding: the non-representational escape). *Let h be a gate whose verdict is determined by a ground-truth oracle for φ on a covered scope $S \subseteq \mathcal{C}$ —for instance, execution of the candidate against a real environment that realizes φ on S —and arbitrary off S . Then $\mathcal{B}(h) \cap S = \emptyset$: h rejects every fault in S , including faults in Σ_F . Composing h with an F -stack lowers the escape floor on S from $\Sigma_F \cap S$ to $\emptyset$, leaving floor $\mu(\Sigma_F \setminus S)$.*

Proof. On S the verdict of h equals φ , so h rejects every $c \in \mathcal{C}_0 \cap S$ and $\mathcal{B}(h) \cap S = \emptyset$; in particular $\Sigma_F \cap S \subseteq \mathcal{C}_0 \cap S$ is rejected by h irrespective of $\sim_F$. The composed stack accepts a fault iff both accept it, so its escaping set is $(\bigcap_g \mathcal{B}(g)) \cap \mathcal{B}(h)$, which on S is empty and off S is at most Σ_F ; the residual floor is the un-grounded stealth $\mu(\Sigma_F \setminus S)$. $\square$

Reality-grounding is the one escape that is not itself a family: its discriminating power comes from φ directly, not from any $\sim$. It is the recovery-block acceptance test of classical fault tolerance, here identified as the unique channel that imports distinctions no representation supplies. Its reach is exactly its grounded scope S ; the faults it cannot escape are those that do not manifest under the available grounding, which is the genuine specification-bound residue.

Remark 4 (Decomposition of the stealth mass; relation to the horizon bound). Write $\Sigma_\cap = \bigcap_i \Sigma_i$ for the intersection of the stealth sets of all families *available* to a system. Then a single family's stealth decomposes as $\Sigma_F = (\Sigma_F \setminus \Sigma_\cap) \uplus \Sigma_\cap$, matching the split of the stealth mass $\lambda_{\text{st}} = \lambda_{\text{red}} + \lambda_{\text{irr}}$ used in the long-horizon bound [17]: the *diversity-reducible* part $\lambda_{\text{red}} = \mu(\Sigma_F \setminus \Sigma_\cap)$ is removed by importing exogenous families (Proposition 1), and the *irreducible* part $\lambda_{\text{irr}} = \mu(\Sigma_\cap)$ —faults every available family confuses—is lowered only by reality-grounding (Proposition 2), down to the specification-bound residue no grounding covers. The companion result then reads: the reliable horizon is capped at $\approx 1/\lambda_{\text{st}}$, and the present paper says λ_{st} is a single founding family's λ_F for a self-generated stack and falls only as exogenous families and grounding are added.

6 Numerical Illustration

We illustrate the floor and its escapes in a transparent fault-population model (full code in the appendix). A founding family F has stealth mass $\lambda_F = 0.15$: a 0.15 fraction of faults lie in Σ_F

and are invisible to every F -gate. On the remaining *visible* faults each F -gate fires independently with probability $q = 0.6$. We measure the escape mass of (i) a tower of h same-family gates, (ii) three different-prompt F -gates, (iii) the F -tower augmented with one exogenous family- N gate of independent stealth λ_N , and (iv) the F -tower augmented with a reality-grounded gate covering a fraction g of the stealth faults. Each figure is a Monte-Carlo average over 2×10^6 faults; reported values match the closed forms to sampling error. Arm (iii) models the theoretical ideal of *independent* blind spots (Proposition 1); as independence is the strong hypothesis flagged in §5, the appendix also reports a correlated variant in which a shared cross-family bias $\rho_{FN} > 0$ raises the floor strictly between $\lambda_F \lambda_N$ and $\min(\lambda_F, \lambda_N)$ —the regime a physical deployment occupies.

configuration	escape mass	closed form
F -tower, depth $h = 1$	0.490	$\lambda_F + (1 - \lambda_F)(1 - q)^h$
F -tower, depth $h = 3$	0.205	$\rightarrow \lambda_F$
F -tower, depth $h = 8$	0.151	$\rightarrow \lambda_F$
F -tower, depth $h = 15$	0.150	$\lambda_F = 0.15$
three different-prompt F -gates	0.205	floored at λ_F (prompt $\neq$ representation)
F -tower + one exogenous N -gate ($\lambda_N = 0.15$)	0.0225	$\lambda_F \lambda_N = 0.0225$
F -tower + reality-grounded gate ($g = 0.5$)	0.075	$\lambda_F(1 - g)$
F -tower + reality-grounded gate ($g = 0.9$)	0.015	$\lambda_F(1 - g)$

Table 1: The same-family tower plateaus at the founding stealth mass $\lambda_F = 0.15$ by depth ≈ 8 and never falls below it (Theorem 2); prompt variation does not move it; only an exogenous family (multiplying to $\lambda_F \lambda_N$, Proposition 1) or a reality-grounded gate (removing the grounded fraction, Proposition 2) breaks the floor.

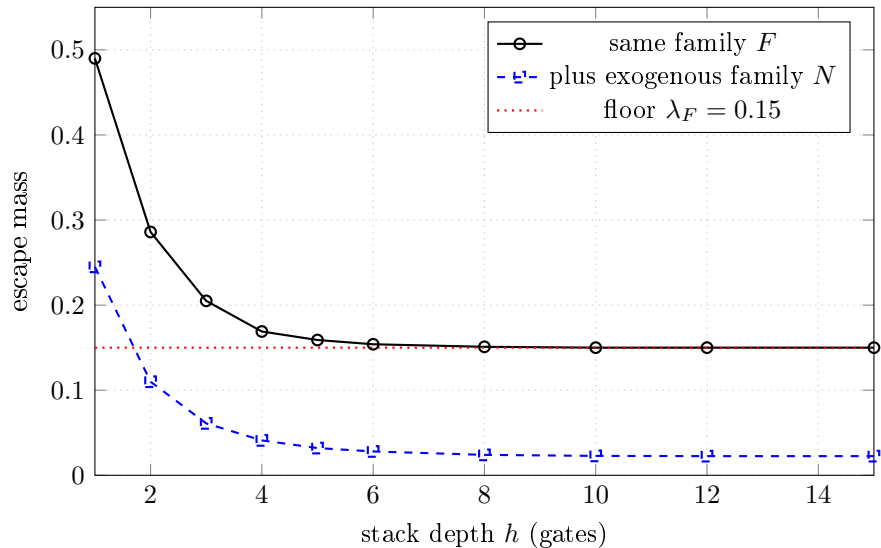


Figure 1: Escape mass versus stack depth. A same-family stack (solid) converges to the founding family’s stealth mass λ_F and stops; adding an exogenous family (dashed) moves the asymptote to $\lambda_F \lambda_N$. No amount of within-family depth crosses the red floor.

7 Discussion

What the floor does and does not say. The floor is about a family’s blind set, not about model quality. A stronger model has a finer $\sim_F$ and hence a smaller Σ_F , so capability does shift the floor; what it cannot do is remove the floor, and within a fixed family no amount of generation reaches below it. The result is therefore orthogonal to the capability trend: better executors lower the founding floor, but only exogenous diversity or grounding crosses it, and a self-generating system that lacks both is capped at its founding family’s stealth regardless of how capable that family is. This is the precise sense in which “let the model check itself” has a ceiling that scaling does not lift.

Consequences for self-improving verification. A system that improves its own checkers by generating new ones from its founding family improves coverage of the faults it can already see and leaves Σ_F untouched; its escape floor is stationary under self-improvement of the gate stack (Theorem 2). Lowering the floor requires adding a distinction source the system does not already contain—another family, or reality-grounding—and such a source must come from outside the self-generation loop, since by inheritance anything the loop produces lies within Σ_F . A design that economizes by replacing an exogenous checker with a self-generated one of the founding family therefore raises the floor back to λ_F while appearing to preserve the gate count: the loss is invisible to any measurement that does not track family provenance. This floor is orthogonal to the *collusion* failure studied in AI control [14], where an untrusted same-family monitor may *intentionally* pass a fault: the stealth floor binds even a perfectly honest monitor, since the blind spot is a limit of representation, not of intent—and the two compound, a same-family checker being both blind and, if misaligned, collusive.

What must be measured. The floor is quantitative only once Σ_F and the cross-family intersection $\Sigma_\cap$ are measured. The blind set is not directly observable, but its mass is estimable from disagreement statistics: faults that one family accepts and an exogenous family rejects bound $\Sigma_F \setminus \Sigma_N$ from below, and faults all families accept that a ground-truth probe later rejects estimate $\Sigma_\cap$. Calibrating the pairwise stealth overlap as a function of representational distance between families, and certifying that a candidate set of families has small joint stealth, are the empirical companions to the theory; we leave their methodology to separate work and note only that the theory tells one exactly which quantity—joint stealth across provenance-independent families—must be driven down.

Limits of the account. Two assumptions bound the result. First, the model treats a gate’s discriminating power as a function of its family’s representation; gates that combine a family with a tool or oracle are covered by Propositions 1–2 but a gate with idiosyncratic, non-family structure is outside the model. Second, the stealth set is defined relative to a fixed correctness predicate φ ; where φ is itself uncertain—the genuine specification gap—the “faults” a family confuses with acceptable work are partly a property of the unresolved specification, and the irreducible residue of Remark 4 inherits that uncertainty. Neither assumption affects the two theorems, which are statements about a fixed φ and a fixed family relation; they bound the *interpretation* of Σ_F , not its consequences.

8 Conclusion

Diverse verification works because gates miss different faults; the question this paper settles is where that diversity can come from. Within a single model family it cannot be generated: a gate built from a family inherits the family's blind set (GG1), and a system that generates its own gates from one founding family is capped at that family's stealth mass, a fixed point no depth, recursion, or prompt variation lowers (GG2). The cap falls only when a distinction the family lacks is imported—a genuinely different family, whose independent blind spot multiplies the floor down, or a reality-grounded gate, whose verdict comes from ground truth rather than representation and which is the unique non-family escape. The classical theory of coincident failures established that independent development does not buy independent failure and that forced diversity helps; we have shown, for the regime where the versions and their checkers are produced from within one family, that the diversity forced diversity requires cannot be supplied from inside, and we have identified the two exogenous sources that supply it. The practical reading is the one the recent empirical record keeps arriving at: a system cannot verify its way out of its own blind spots, and what it needs is not more of itself but something genuinely other—another family, or contact with reality.

A Simulation code

The following self-contained script reproduces Table 1 and the asymptotes of Figure 1. It models faults as stealth-to- F (a fraction λ_F , invisible to every F -gate) or visible (each F -gate fires with probability q), and evaluates the same-family tower, the prompt-diversity arm, the exogenous-family arm (independent and correlated), and the reality-grounded arm against their closed forms.

```
import numpy as np
rng = np.random.default_rng(7)

N = 2_000_000 # faults sampled
lambda_F = 0.15 # founding-family stealth mass
q = 0.6 # per-F-gate catch prob on a VISIBLE fault

stealth_F = rng.random(N) < lambda_F # invisible to every F-gate

def F_tower_escape(h):
    """escape mass after h independent same-family-F gates."""
    if h == 0:
        return 1.0
    missed_visible = (rng.random((N, h)) > q).all(axis=1)
    return (stealth_F | (~stealth_F & missed_visible)).mean()

print("== same-family tower: plateaus at lambda_F, never below ==")
for h in [1, 2, 3, 5, 8, 15]:
    print(f" h={h:2d} escape={F_tower_escape(h):.4f} (floor lambda_F={lambda_F})")

print("\n== prompt diversity within F: stealth set is a property of the")
print(" representation, not the prompt -> floor unchanged ==")
mv = (rng.random((N, 3)) > q).all(axis=1)
print(f" 3 different-prompt F-gates escape={(stealth_F | (~stealth_F & mv)).mean():.4f}")

print("\n== add ONE exogenous family-N gate (independent blind spot) ==")
for lambda_N in [0.15, 0.30]:
    stealth_N = rng.random(N) < lambda_N
    # big F-stack rejects all visible-to-F faults; survivors are stealth-to-F,
    # of which the N-gate rejects those it can see -> stealth-to-both remains
    print(f" lambda_N={lambda_N}: escape={(stealth_F & stealth_N).mean():.4f}")
    print(f" (predicted lambda_F*lambda_N={lambda_F*lambda_N:.4f})")

print("\n== exogenous N with CORRELATION rho: independence (rho=0) is the ideal;")
print(" shared cross-family bias raises the floor toward min(1F,1N) ==")
```

```

lambda_N = 0.15
sd = (lambda_F*(1-lambda_F)*lambda_N*(1-lambda_N))*0.5
for rho in [0.0, 0.5, 1.0]:
    p_both = lambda_F*lambda_N + rho*sd          # target P(stealth in BOTH families)
    u = rng.random(N)
    stealth_N = np.where(stealth_F, u < p_both/lambda_F,
                        u < (lambda_N - p_both)/(1 - lambda_F))
    print(f" rho={rho}: escape={{(stealth_F & stealth_N).mean():.4f}}
          f" (1F*1N={{lambda_F*lambda_N:.4f}} min={{min(lambda_F,lambda_N):.4f}})")

print("\n== reality-grounded gate: rejects grounded fraction g of stealth faults ==")
for g in [0.0, 0.5, 0.9]:
    grounded_catch = rng.random(N) < g
    print(f" g={g}: escape={{(stealth_F & ~grounded_catch).mean():.4f}}
          f" (predicted lambda_F*(1-g)={{lambda_F*(1-g):.4f}})")

```

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